

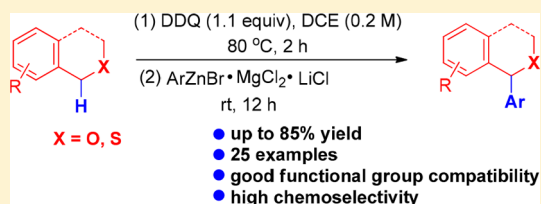
Direct Arylation of Benzyl Ethers with Organozinc Reagents

Zhihua Peng,*¹ Yilei Wang, Zhi Yu, Dezhi Zhao, Linhua Song, and Cuiyu Jiang

Department of Chemistry, College of Science, China University of Petroleum (East China), Qingdao, Shandong 266580, P. R. China

Supporting Information

ABSTRACT: A novel C(sp³)–H bond arylation of benzyl ethers with Knochel-type arylzinc reagents has been developed. This transition-metal-catalyst-free reaction proceeds well under mild conditions in a simple and effective manner and enables the synthesis of a wide range of potentially biologically active benzyl ethers by using highly functionalized organozinc reagents as a carbon nucleophile.



INTRODUCTION

Functionalized benzyl ethers such as isochroman are common structural motifs in biologically active natural compounds and pharmaceuticals.¹ In particular, 1-arylated benzyl ethers not only possess significant antioxidants, antiviral, and anticancer activities² but also have been found in widespread applications as an important building block in the synthesis of drug molecules.³ Considering the importance and widespread applications of 1-arylated benzyl ethers, the synthesis of these compounds therefore has been an ongoing challenge and is a promising area of research. In the past few decades, the selective functionalization of reactive C(sp³)–H bonds at the C1 position of benzyl ethers has become the most attractive methodology for the synthesis of 1-functionalized benzyl ethers as it may represent a step- and atom-economic strategy.⁴ In this context, cross-dehydrogenative coupling (CDC) reactions⁵ have been developed as an effective method for the construction of α -substituted benzyl ethers. Although a few oxidation systems and CDCs between benzyl ethers and a series of carbon nucleophiles have been studied,⁶ much fewer methods for the synthesis of 1-arylated benzyl ethers were reported⁷ because performing a CDC reaction between benzyl ethers and aromatic rings acting as nucleophiles is still a challenge. Alternatively, the reaction of organometallic reagents with preoxidized benzyl ethers offered an effective approach for the synthesis of 1-arylated benzyl ethers. The advantages of this kind of C(sp³)–H bond functionalization of benzyl ethers with organometallic reagents includes transition-metal-free and mild reaction conditions and a high regioselectivity. Muramatsu reported several oxidation systems to achieve a one-pot oxidative coupling of isochroman with Grignard reagents, affording diverse α -substituted derivatives in a high efficiency.⁸ In addition, Liu's group has disclosed a trityl ion-mediated practical C–H functionalization of benzopyrans with a wide range of nucleophiles (organoboranes and C–H molecules) at ambient temperature.⁹ Despite the fact that the great innovation has been achieved, some disadvantages such as unfavorable regioselectivity and poor functional group tolerance might be brought by the used organometallic reagents, which may limit the broad application of this

protocol in organic synthesis. To overcome the limitation, the employment of highly functionalized organometallic reagents with a better selectivity and reactivity is crucial in this transformation. Among various organometallics, organozinc reagents exhibit better compatibility with sensitive functional groups, and they are one of the most versatile organometallic reagents for the synthesis of complex compounds.¹⁰ To date, most of the C(sp³)–H bond functionalizations with organozinc reagents mainly involve oxidative couplings next to nitrogen atoms. The group of Menche revealed CuCl₂-catalyzed alkylation and benzylation of THIQ derivatives with organozinc reagents.¹¹ In 2017, our group has demonstrated a novel transition-metal-catalyst-free C(sp³)–H bond arylation of tetrahydroisoquinoline (THIQ) derivatives with Knochel-type arylzinc reagents.¹² However, the reports on the reaction of the corresponding less reactive C(sp³)–H bonds next to an oxygen atom with less reactive organozinc reagents are scarce. One case was reported by Muramatsu in which Ph₂Zn was used in the catalytic C(sp³)–H bond arylation of isochroman during the optimization process. In view of the synthetic utility and certain limitations of existing methods in particular regarding the substrate scope, herein, we describe a convenient, transition-metal-catalyst-free and broadly applicable oxidative cross-coupling reaction of benzyl ethers with arylzinc reagents to access diverse 1-arylated benzyl ether derivatives in an effective manner.

RESULTS AND DISCUSSION

According to the analogous study in our group, the phenylzinc reagent coordinated with MgCl₂ and LiCl (PhZnBr·MgCl₂·LiCl)¹³ shows an increasing reactivity to the iminium cation formed by a nucleophilic substitution of tetrahydroisoquinoline (THIQ) to [bis(trifluoroacetoxy)iodo]benzene (PIFA) and deprotonation of the generated ammonium cation with the trifluoroacetate ion because of the presence of MgCl₂ and LiCl.¹² We assumed that this kind of organozinc reagent might also display a better reactivity to the oxonium cation generated

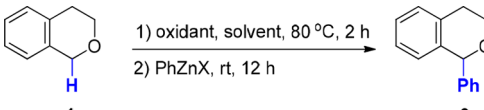
Received: March 28, 2018

Published: June 12, 2018



by the oxidation of benzyl ethers with organic oxidants. Therefore, we selected the coupling of isochroman with $\text{PhZnBr}\cdot\text{MgCl}_2\cdot\text{LiCl}$ as a starting point for our initial study. The oxidation of isochroman to an oxonium ion could be achieved by using 2,3-dicyano-5,6-dichlorobenzoquinone (DDQ) as an organic oxidant in PhCl at 80 °C within 2 h.^{8a} The treatment of this intermediate ion with $\text{PhZnBr}\cdot\text{MgCl}_2\cdot\text{LiCl}$, prepared by the insertion of magnesium turnings into phenyl bromide in the presence of ZnCl_2 and LiCl, afforded the desired product **3a** in the yield of 77% (Table 1, entry 1).

Table 1. Optimization of Reaction Conditions^a

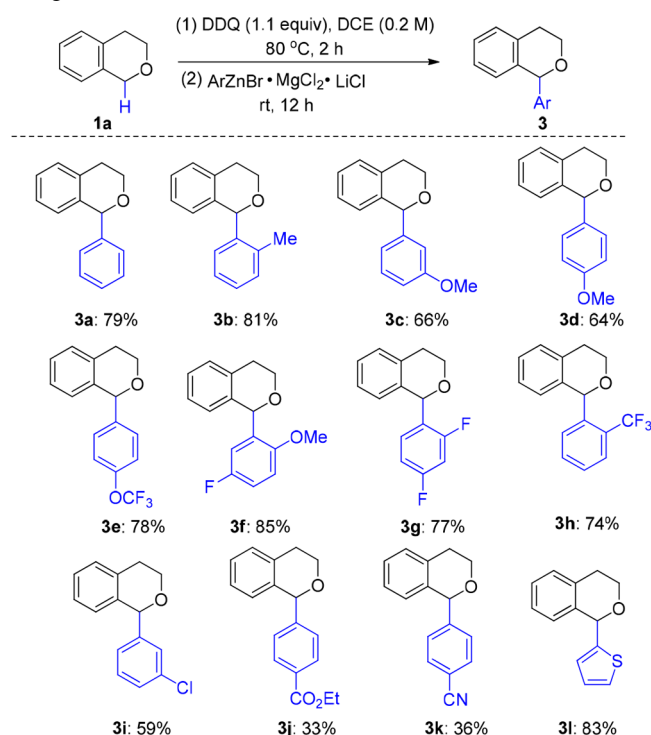
				
entry	oxidant	solvent	PhZnX	yield of 3a ^b (%)
1	DDQ ^c	PhCl	$\text{PhZnBr}\cdot\text{MgCl}_2\cdot\text{LiCl}$	77
2	DDQ ^c	PhCl	$\text{PhZnI}\cdot\text{MgCl}_2\cdot\text{LiCl}$	70
3	DDQ ^c	PhCl	$\text{PhZnI}\cdot\text{LiCl}$	16
4	DDQ ^c	DCE	$\text{PhZnBr}\cdot\text{MgCl}_2\cdot\text{LiCl}$	79
5	DDQ ^c	THF	$\text{PhZnBr}\cdot\text{MgCl}_2\cdot\text{LiCl}$	trace
6	3,3',5,5'-tetra- <i>tert</i> -butyldiphenylquinone ^c	DCE	$\text{PhZnBr}\cdot\text{MgCl}_2\cdot\text{LiCl}$	trace
7	3,3',5,5'-tetra- <i>tert</i> -butyldiphenylquinone, PIFA ^d	DCE	$\text{PhZnBr}\cdot\text{MgCl}_2\cdot\text{LiCl}$	trace
8	DEAD ^e	THF	$\text{PhZnBr}\cdot\text{MgCl}_2\cdot\text{LiCl}$	0
9	DEAD, PIFA ^f	THF	$\text{PhZnBr}\cdot\text{MgCl}_2\cdot\text{LiCl}$	0

^aGeneral conditions: after the mixture of **1a** (0.5 mmol) and oxidant (0.2 to 1.1 equiv) was stirred for 2 h in solvent (2.5 mL) at room temperature to 80 °C, organozinc reagent (1.5 mmol) was added, and the reaction mixture was stirred for 12 h at room temperature. ^bIsolated yield. ^c1.1 equiv of oxidant was used. ^dPIFA (1.1 equiv) and 3,3',5,5'-tetra-*tert*-butyldiphenylquinone (20 mol %) were used. ^eThe oxidation of **1a** (0.5 mmol) with DEAD (1.1 equiv) was conducted at room temperature within 5 h. ^fThe oxidation of **1a** (0.5 mmol) with DEAD (20 mol %) and PIFA (1.1 equiv) was conducted at room temperature within 5 h.

The use of $\text{PhZnI}\cdot\text{MgCl}_2\cdot\text{LiCl}$, prepared by the insertion of magnesium turnings into phenyl iodide in the presence of ZnCl_2 and LiCl, slightly decreased the yield to 70% (Table 1, entry 2). Unsurprisingly, the oxidation of isochroman (**1a**) in the presence of DDQ in PhCl and subsequent reaction with $\text{PhZnI}\cdot\text{LiCl}$ ¹⁴ (2 equiv) provided the corresponding product **3a** in the isolated yield of 16% (Table 1, entry 3). In 1,2-dichloroethane (DCE), the reaction proceeded well to afford **3a** in a 79% yield (Table 1, entry 4). However, the usage of THF as a solvent did not give a satisfactory result (Table 1, entry 5). Oxidant screening was also carried out. When oxidants such as 3,3',5,5'-tetra-*tert*-butyldiphenylquinone, diethyl azodicarboxylate (DEAD), or their mixtures with PIFA were used instead of DDQ, only a trace amount of coupling product **3a** was obtained or no desired product was observed (Table 1, entries 6–9).

Having established the optimized reaction conditions, we tested the substituent effect of arylzinc reagents on this transformation (Scheme 1). The reaction of arylzinc reagents bearing electron-donating groups with isochroman proceeded well, providing the desired products in good to excellent yields. The oxidation of isochroman (0.5 mmol) with DDQ (1.1

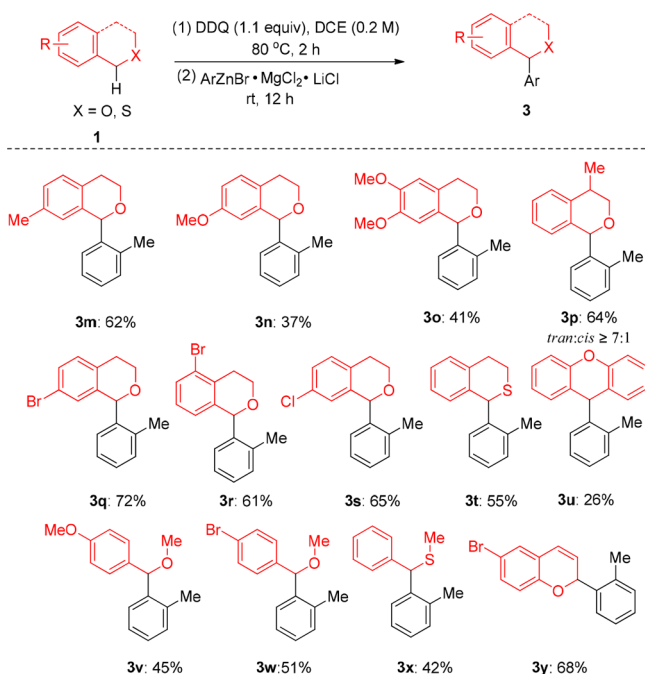
Scheme 1. Coupling of Isochroman with Various Arylzinc Reagents^{a,b}



^aThe reaction was performed by the oxidation of **1a** (0.5 mmol) and DDQ (1.1 equiv) in DCE (2.5 mL) at 80 °C for 2 h and subsequent treatment with organozinc reagent (1.5 mmol) at room temperature for 12 h. ^bIsolated yield.

equiv) in DCE (2.5 mL) at 80 °C within 2 h and subsequent treatment with 2-methylphenylzinc reagent prepared by the insertion of magnesium into 1-bromo-2-methylbenzene in the presence of ZnCl_2 and LiCl afforded the coupling product **3b** in the yield of 81%. Under similar conditions, the oxidative cross-coupling of arylzinc reagents bearing a methoxy group with isochroman gave the corresponding products **3c–d** in 64–66% yields. In addition, compound **3e** containing a trifluoromethoxy group was produced in the yield of 78% in a similar manner. Arylzinc reagents containing electron-withdrawing groups could also react with isochroman, producing the desired products **3f–i** in good yields. Notably, the oxidative arylation reaction of isochroman with arylzinc reagents could tolerate an ester group. The reaction of isochroman with (4-(ethoxycarbonyl)phenyl)zinc reagent under similar conditions delivered the coupling product **3j** in 33% yield. Interestingly, arylzinc reagent containing a cyano group reacted well with isochroman, affording the expected product **3k** in 36% yield. Additionally, the heterocyclic compound **3l** could also be obtained in the yield of 83%. It is worth mentioning that a variety of functional groups such as OMe (**3c**, **3d**, **3f**), OCF_3 (**3e**), F (**3f**, **3g**), CF_3 (**3h**), Cl (**3i**), CO_2Et (**3j**), and CN (**3k**) were tolerated well under the reaction conditions. However, the reaction of arylzinc reagents containing azide, amide, pyridinyl, or carbonyl groups under the optimum conditions did not provide the desired products.

We next studied the generality of benzyl ethers (Scheme 2). In a similar manner, isochroman bearing a Me group at the C7 position reacted successfully with 2-MeC₆H₄ZnBr·MgCl₂·LiCl, affording the corresponding coupled product **3m** in the yield of

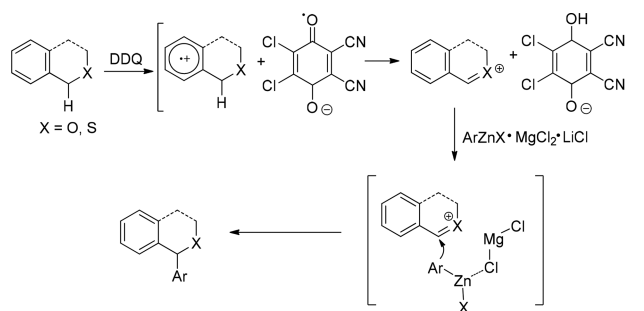
Scheme 2. Scope of Benzyl Ethers in the C–H Bond Arylation^{a,b}

^aThe reaction was performed by the oxidation of **1** (1 mmol) and DDQ (1.1 equiv) in DCE (5 mL) at 80 °C for 2 h and subsequent treatment with organozinc reagent (3 mmol) at room temperature for 12 h. ^bIsolated yield.

62%. Gratifyingly, the treatment of 2-methylphenyl zinc reagent with 7-methoxyisochroman or 6,7-dimethoxyisochroman, which are the less reactive coupling partners due to the presence of MeO-group at the C6 or C7 position of isochroman, furnished the desired coupling products **3n–o** in 37–41% yields. In addition, 4-methylisochroman coupled smoothly with 2-MeC₆H₄ZnBr·MgCl₂·LiCl, leading to the arylated product **3p** in 64% yield with a good selectivity (*trans/cis* ≥ 7:1). The C(sp³)–H bond arylation of benzyl ethers with arylzinc reagents displays a good compatibility with a halide substituent. The reaction went smoothly when a bromo substituent was placed at the C7 or C5 position, and the coupled products **3q–r** were produced in 61–72% yields. Under identical conditions, the reaction of isochroman bearing a chlorine group at the C7 position with 2-MeC₆H₄ZnBr·MgCl₂·LiCl provided the corresponding coupled product **3s** in the yield of 65%. In addition, isothiochroman reacted well with 2-MeC₆H₄ZnBr·MgCl₂·LiCl under the standard conditions, giving the isothiochroman derivative **3t** in the yield of 55%. Xanthene also reacted smoothly with 2-methylphenyl zinc reagent under the standard conditions and afforded the desired product **3u** in 26% yield. Acyclic benzyl ethers such as 1-methoxy-4-(methoxymethyl)benzene, 1-bromo-4-(methoxymethyl)benzene and benzyl methyl sulfide were also compatible with this coupling system, though the expected products **3v–x** were obtained only in moderate yields. 6-Bromo-2*H*-chromene also proved to be a competent substrate. The reaction of 6-bromo-2*H*-chromene with 2-MeC₆H₄ZnBr·MgCl₂·LiCl led to the expected product **3y** in 68% yield.

A plausible reaction mechanism for the C(sp³)–H arylation of benzyl ethers with arylzinc reagents is shown in Scheme 3. A radical cation generated by a single electron transfer from

Scheme 3. Proposed Mechanism for the C–H Bond Arylation of Benzyl Ethers with Arylzinc Reagents



benzyl ethers to DDQ is accessed through H atom abstraction to form an oxonium cation.¹⁵ The subsequent addition of reactive arylzinc reagents to oxonium cation affords the desired coupling products.

CONCLUSION

In summary, we have developed a novel C(sp³)–H bond arylation of benzyl ethers derivatives with Knochel-type arylzinc reagents. The reactions could be readily performed under mild conditions without the use of a heavy metal catalyst, affording a wide range of potentially biologically active compounds. In addition, this transformation exhibits excellent compatibility with various sensitive functional groups such as an ester group. Further studies on the scope and applications of this reaction are now under investigation.

EXPERIMENTAL SECTION

General Information. All reactions were carried out under a nitrogen atmosphere in flame-dried glassware. Syringes, which were used to transfer anhydrous solvents or reagents, were purged with nitrogen prior to use. THF was continuously refluxed and freshly distilled from sodium benzophenone ketyl under nitrogen. Yields refer to isolated yields of compounds estimated to be >95% pure as determined by ¹H NMR (25 °C) and capillary-GC. NMR spectra were recorded on solutions in deuterated chloroform (CDCl₃) with residual chloroform (δ 7.25 ppm for ¹H NMR and δ 77.0 ppm for ¹³C NMR). Abbreviations for signal coupling are as follows: s, singlet; d, doublet; t, triplet; q, quartet; m, multiplet; br, broad. ESI-QTOF-MS measurements were performed in the positive ion mode (*m/z* 50–2000). Column chromatography purifications were performed using SiO₂ (200–300 mesh ASTM) from Branch of Qingdao Haiyang Chemical Co., Ltd. if not indicated otherwise. Magnesium turnings, zinc chloride anhydrous (98%), and lithium chloride (AR) were purchased from Sinopharm Chemical Reagent Co., Ltd.

Typical Procedure for the Preparation of Arylzinc Reagents.

A dry and nitrogen-flushed Schlenk flask, equipped with a magnetic stirrer and a septum, was charged with LiCl (191 mg, 4.5 mmol) and was heated under a vacuum until dry. ZnCl₂ (450 mg, 3.3 mmol) was added and also heated under a vacuum until dry. Magnesium powder (180 mg, 7.5 mmol) was added and heated under a vacuum until dry. Dry THF (6 mL) was added, and the mixture was stirred for 30 min at room temperature. Then, aryl halide (3.0 mmol) was added in one portion at 0–25 °C. The reaction mixture was stirred for 2–12 h, affording the corresponding arylzinc reagents.

Typical Procedure (TP1) for the Reaction of Benzyl Ethers with Arylzinc Reagents.

A clean Schlenk tube was dried for 5 min at 380 °C (heat gun) under a high vacuum (1 mbar). After cooling to room temperature, the tube was evacuated and backfilled with nitrogen three times. To the mixture of DDQ (1.1 equiv) and 1,2-dichloroethane (0.2 M) was added benzyl ether (1 equiv). The reaction mixture was stirred for 2 h at 80 °C. The mixture was allowed to cool to room temperature, and the corresponding aromatic zinc

reagent (3 equiv) was added dropwise at room temperature. After the mixture was stirred for 12–16 h, water (20 mL) was then added to the reaction mixture. The aqueous layer was extracted with ethyl acetate (3 × 20 mL). The combined organic phases were washed with brine and dried over magnesium sulfate. The solvent was removed by rotary evaporation, and purification by flash column chromatography on silica gel using petroleum ether/ethyl acetate as an eluent gave the expected products.

1-Phenylisochroman (3a). According to **TP1**, the reaction of isochroman (67 mg, 0.5 mmol) with phenylzinc reagent (4.5 mL, 0.33 M) afforded the desired product **3a** which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:50) as a white solid (83 mg, 79%): mp 85–87 °C (ethyl ether/petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ (ppm) 7.24–7.14 (m, 5H), 7.06–6.99 (m, 2H), 6.96–6.89 (m, 1H), 6.63 (d, *J* = 7.8 Hz, 1H), 5.60 (s, 1H), 4.06 (ddd, *J* = 11.3, 5.6, 3.9 Hz, 1H), 3.79 (ddd, *J* = 11.3, 9.5, 3.9 Hz, 1H), 3.00 (ddd, *J* = 16.3, 9.5, 5.6 Hz, 1H), 2.66 (dt, *J* = 16.3, 3.9 Hz, 1H); ¹³C NMR (100 MHz, CDCl₃) δ (ppm) 142.1, 137.2, 133.7, 128.8, 128.3, 127.9, 126.8, 126.5, 125.8, 79.5, 63.7, 28.7; IR (KBr) 3151, 2964, 2873, 1599, 1489, 1452, 1281, 1100, 904 cm⁻¹; HRMS (ESI) calcd for C₁₅H₁₄NaO [M + Na]⁺ 233.0942, found 233.0936; *R*_f 0.50 (petroleum ether/EtOAc, 20:1).

1-(*o*-Tolyl)isochroman (3b). According to **TP1**, the reaction of isochroman (67 mg, 0.5 mmol) with 2-methylphenylzinc reagent (3.3 mL, 0.46 M) afforded the desired product **3b** which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:50) as a pale yellow oil (92 mg, 81%). ¹H NMR (400 MHz, CDCl₃) δ (ppm) 7.15–6.92 (m, 7H), 6.60 (d, *J* = 7.6 Hz, 1H), 5.83 (s, 1H), 4.11 (ddd, *J* = 11.3, 5.5, 3.7 Hz, 1H), 3.84 (ddd, *J* = 11.3, 9.7, 3.9 Hz, 1H), 3.05 (ddd, *J* = 15.9, 9.7, 5.6 Hz, 1H), 2.66 (dt, *J* = 16.3, 3.7 Hz, 1H), 2.26 (s, 3H); ¹³C NMR (100 MHz, CDCl₃) δ (ppm) 139.9, 137.5, 137.2, 133.9, 130.9, 129.9, 128.6, 128.0, 126.5, 126.3, 125.9, 125.7, 77.7, 64.0, 28.8, 19.3; IR (KBr) 3064, 3021, 2927, 1489, 1257, 1088 cm⁻¹; HRMS (ESI) calcd for C₁₆H₁₆NaO [M + Na]⁺ 247.1099, found 247.1090; *R*_f 0.54 (petroleum ether/EtOAc, 20:1).

1-(3-Methoxyphenyl)isochroman (3c). According to **TP1**, the reaction of isochroman (67 mg, 0.5 mmol) with 3-methoxyphenylzinc reagent (3.8 mL, 0.40 M) afforded the desired product **3c**, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:30) as a colorless oil (79 mg, 66%): ¹H NMR (400 MHz, CDCl₃) δ (ppm) 7.20–6.95 (m, 4H), 6.81 (d, *J* = 7.6 Hz, 1H), 6.79–6.73 (m, 2H), 6.69 (d, *J* = 7.6 Hz, 1H), 5.61 (s, 1H), 4.10 (ddd, *J* = 11.3, 5.4, 4.2 Hz, 1H), 3.83 (ddd, *J* = 11.3, 9.5, 3.9 Hz, 1H), 3.67 (s, 3H), 3.03 (ddd, *J* = 15.8, 9.6, 5.6 Hz, 1H), 2.70 (dt, *J* = 16.3, 3.7 Hz, 1H); ¹³C NMR (100 MHz, CDCl₃) δ (ppm) 159.6, 143.6, 137.1, 133.7, 129.3, 128.6, 126.8, 126.6, 125.8, 121.2, 114.2, 113.6, 79.5, 63.8, 55.1, 28.7; IR (KBr) 3151, 2929, 2851, 1665, 1578, 1124 cm⁻¹; HRMS (ESI) calcd for C₁₆H₁₆NaO₂ [M + Na]⁺ 263.1048, found 263.1038; *R*_f 0.48 (petroleum ether/EtOAc, 10:1).

1-(4-Methoxyphenyl)isochroman (3d). According to **TP1**, the reaction of isochroman (67 mg, 0.5 mmol) with 4-methoxyphenylzinc reagent (3.7 mL, 0.41 M) afforded the desired product **3d**, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:30) as a white solid (77 mg, 64%): mp 72–74 °C (ethyl ether/petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ (ppm) 7.13 (d, *J* = 8.6 Hz, 2H), 7.08–7.04 (m, 2H), 7.01–6.95 (m, 1H), 6.78 (d, *J* = 8.6 Hz, 2H), 6.66 (d, *J* = 7.6 Hz, 1H), 5.60 (s, 1H), 4.08 (ddd, *J* = 11.3, 5.4, 3.9 Hz, 1H), 3.82 (ddd, *J* = 11.3, 9.4, 3.9 Hz, 1H), 3.69 (s, 3H), 3.03 (ddd, *J* = 16.3, 9.4, 5.4 Hz, 1H), 2.70 (dt, *J* = 16.3, 3.9 Hz, 1H); ¹³C NMR (100 MHz, CDCl₃) δ (ppm) 159.3, 137.6, 134.4, 133.8, 130.0, 128.6, 126.9, 126.5, 125.8, 113.7, 79.0, 63.7, 55.2, 28.8; IR (KBr) 3001, 2949, 2831, 1615, 1517, 1250, 1176 cm⁻¹; HRMS (ESI) calcd for C₁₆H₁₆NaO₂ [M + Na]⁺ 263.1048, found 263.1045; *R*_f 0.47 (petroleum ether/EtOAc, 10:1).

1-(4-(Trifluoromethoxy)phenyl)isochroman (3e). According to **TP1**, the reaction of isochroman (67 mg, 0.5 mmol) with 4-(trifluoromethoxy)phenylzinc reagent (3.2 mL, 0.47 M) afforded the desired product **3e**, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:30) as a pale yellow oil (115 mg, 78%): ¹H NMR (400 MHz, CDCl₃) δ (ppm) 7.27 (d, *J* =

8.6 Hz, 2H), 7.18–7.07 (m, 4H), 7.05–6.98 (m, 1H), 6.65 (d, *J* = 7.8 Hz, 1H), 5.67 (s, 1H), 4.11 (ddd, *J* = 11.4, 5.5, 3.7 Hz, 1H), 3.85 (ddd, *J* = 11.4, 9.6, 3.7 Hz, 1H), 3.07 (ddd, *J* = 16.5, 9.6, 5.5 Hz, 1H), 2.74 (dt, *J* = 16.5, 3.6 Hz, 1H); ¹³C NMR (100 MHz, CDCl₃) δ (ppm) 148.9 (q, *J* = 2.2 Hz), 140.9, 136.7, 133.8, 130.3, 128.9, 126.8, 126.7, 126.0, 120.9, 120.4 (q, *J* = 257.8 Hz), 78.8, 63.9, 28.7; IR (KBr) 3068, 2931, 2858, 1600, 1508, 1261, 1166, 1092 cm⁻¹; HRMS (ESI) calcd for C₁₆H₁₃F₃NaO₂ [M + Na]⁺ 317.0765, found 317.0753; *R*_f 0.45 (petroleum ether/EtOAc, 10:1).

1-(5-Fluoro-2-methoxyphenyl)isochroman (3f). According to **TP1**, the reaction of isochroman (67 mg, 0.5 mmol) with 5-fluoro-2-methoxyphenylzinc reagent (3.3 mL, 0.46 M) afforded the desired product **3f**, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:30) as a colorless oil (110 mg, 85%): ¹H NMR (400 MHz, CDCl₃) δ (ppm) 7.19–7.13 (m, 2H), 7.09–7.04 (m, 1H), 6.95 (ddd, *J* = 9.1, 7.6, 3.2 Hz, 1H), 6.88 (dd, *J* = 9.1, 4.4 Hz, 1H), 6.82 (dd, *J* = 9.1, 3.2 Hz, 1H), 6.75 (d, *J* = 7.6 Hz, 1H), 6.22 (s, 1H), 4.19 (ddd, *J* = 11.4, 5.4, 3.9 Hz, 1H), 3.94 (ddd, *J* = 11.4, 9.4, 3.9 Hz, 1H), 3.87 (s, 3H), 3.18–3.08 (m, 1H), 2.80 (dt, *J* = 16.2, 3.9 Hz, 1H); ¹³C NMR (100 MHz, CDCl₃) δ (ppm) 157.0 (d, *J* = 239.1 Hz), 153.5 (d, *J* = 2.2 Hz), 137.1, 133.9, 132.6 (d, *J* = 6.6 Hz), 128.7, 126.5, 126.4, 126.0, 116.5 (d, *J* = 23.5 Hz), 115.1 (d, *J* = 23.4 Hz), 111.9 (d, *J* = 8.1 Hz), 72.2 (d, *J* = 1.5 Hz), 63.9, 56.3, 28.8; IR (KBr) 2962, 2925, 2854, 1776, 1591, 1558, 1361, 1187, 1028 cm⁻¹; HRMS (ESI) calcd for C₁₆H₁₆FO₂ [M + H]⁺ 259.1134, found 259.1124; *R*_f 0.44 (petroleum ether/EtOAc, 10:1).

1-(2,4-Difluorophenyl)isochroman (3g). According to **TP1**, the reaction of isochroman (67 mg, 0.5 mmol) with (2,4-difluorophenyl) zinc reagent (3.2 mL, 0.47 M) afforded the desired product **3g**, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:30) as a colorless oil (95 mg, 77%): ¹H NMR (400 MHz, CDCl₃) δ (ppm) 7.13–6.95 (m, 4H), 6.80–6.68 (m, 2H), 6.64 (d, *J* = 7.6 Hz, 1H), 5.97 (s, 1H), 4.09 (ddd, *J* = 11.4, 5.6, 3.6 Hz, 1H), 3.84 (ddd, *J* = 11.4, 9.8, 3.9 Hz, 1H), 3.10–3.00 (m, 1H), 2.70 (dt, *J* = 16.4, 3.6 Hz, 1H); ¹³C NMR (100 MHz, CDCl₃) δ (ppm) 162.6 (dd, *J* = 249.4, 11.7 Hz), 161.1 (dd, *J* = 250.9, 11.7 Hz), 136.4, 134.0, 131.4 (dd, *J* = 9.9, 5.5 Hz), 128.8, 126.8, 126.2, 126.1, 125.6 (dd, *J* = 13.2, 3.7 Hz), 111.3 (dd, *J* = 21.3, 3.7 Hz), 103.8 (t, *J* = 24.9 Hz), 72.2 (d, *J* = 2.9 Hz), 64.1, 28.7; IR (KBr) 3078, 2928, 2856, 1664, 1609, 1502, 1425, 1271, 1141 cm⁻¹; HRMS (ESI) calcd for C₁₅H₁₂F₂NaO [M + Na]⁺ 269.0754, found 269.0747; *R*_f 0.23 (petroleum ether/EtOAc, 20:1).

1-(2-(Trifluoromethyl)phenyl)isochroman (3h). According to **TP1**, the reaction of isochroman (67 mg, 0.5 mmol) with (2-(trifluoromethyl)phenyl)zinc reagent (3.3 mL, 0.46 M) afforded the desired product **3h**, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:50) as a pale yellow oil (103 mg, 74%): ¹H NMR (400 MHz, CDCl₃) δ (ppm) 7.71 (d, *J* = 7.8 Hz, 1H), 7.44 (ddd, *J* = 29.2, 7.3, 7.2 Hz, 2H), 7.31 (d, *J* = 7.8 Hz, 1H), 7.17 (d, *J* = 3.9 Hz, 2H), 7.05 (ddd, *J* = 8.0, 4.3, 4.1 Hz, 1H), 6.60 (d, *J* = 7.8 Hz, 1H), 6.14 (s, 1H), 4.29 (ddd, *J* = 11.5, 5.9, 2.5 Hz, 1H), 3.99 (ddd, *J* = 11.2, 11.2, 3.6 Hz, 1H), 3.26 (ddd, *J* = 16.6, 10.7, 5.9 Hz, 1H), 2.84–2.73 (m, 1H); ¹³C NMR (100 MHz, CDCl₃) δ (ppm) 141.2 (q, *J* = 1.5 Hz), 137.6, 133.9, 132.1 (q, *J* = 2.2 Hz), 131.4, 129.1 (q, *J* = 30.1 Hz), 128.7, 127.9, 126.9, 126.8, 126.2, 125.2 (q, *J* = 5.9 Hz), 124.4 (q, *J* = 273.6 Hz), 74.9 (q, *J* = 2.2 Hz), 64.9, 28.8; IR (KBr) 3068, 2952, 2858, 1606, 1261, 1171, 1085 cm⁻¹; HRMS (ESI) calcd for C₁₆H₁₃F₃NaO [M + Na]⁺ 301.0816, found 301.0813; *R*_f 0.54 (petroleum ether/EtOAc, 20:1).

1-(3-Chlorophenyl)isochroman (3i). According to **TP1**, the reaction of isochroman (134 mg, 1 mmol) with (3-chlorophenyl) zinc reagent (8.3 mL, 0.36 M) afforded the desired product **3i**, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:50) as a colorless oil (144 mg, 59%): ¹H NMR (400 MHz, CDCl₃) δ (ppm) 7.31–7.26 (m, 3H), 7.23–7.14 (m, 3H), 7.12–7.07 (m, 1H), 6.74 (d, *J* = 7.8 Hz, 1H), 5.69 (s, 1H), 4.17 (ddd, *J* = 11.4, 5.6, 3.8 Hz, 1H), 3.92 (ddd, *J* = 11.4, 9.5, 4.0 Hz, 1H), 3.19–3.09 (m, 1H), 2.81 (dt, *J* = 16.4, 3.8 Hz, 1H); ¹³C NMR (100 MHz, CDCl₃) δ (ppm) 144.2, 136.5, 134.4, 133.8, 129.6, 128.9, 128.8, 128.3, 127.0, 126.8, 126.7, 126.1, 78.9, 63.9, 28.7; IR (KBr) 3003, 2944, 2293,

1444, 1376, 1196, 1133 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{15}\text{H}_{13}\text{ClNaO}$ $[\text{M} + \text{Na}]^+$ 267.0553, found 267.0542; R_f 0.51 (petroleum ether/EtOAc, 20:1).

Ethyl 4-(isochroman-1-yl)benzoate (3j). According to TP1, the reaction of isochroman (67 mg, 0.5 mmol) with (4-(ethoxycarbonyl)-phenyl)zinc reagent (6.5 mL, 0.23 M) afforded the desired product 3j, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:20) as a colorless oil (47 mg, 33%): ^1H NMR (400 MHz, CDCl_3) δ (ppm) 8.02 (d, J = 8.6 Hz, 2H), 7.38 (d, J = 8.6 Hz, 2H), 7.20–7.15 (m, 2H), 7.09–7.03 (m, 1H), 6.69 (d, J = 7.8 Hz, 1H), 5.76 (s, 1H), 4.37 (q, J = 7.1 Hz, 2H), 4.19 (ddd, J = 11.4, 5.5, 3.9 Hz, 1H), 3.93 (ddd, J = 11.4, 9.5, 4.0 Hz, 1H), 3.19–3.10 (m, 1H), 2.81 (dt, J = 16.4, 3.9 Hz, 1H), 1.38 (t, J = 7.1 Hz, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 166.4, 147.0, 136.7, 133.7, 130.3, 129.7, 128.9, 128.7, 126.9, 126.7, 126.0, 79.1, 64.0, 60.9, 28.7, 14.3; IR (KBr) 3067, 2929, 2853, 1716, 1453, 1407, 1276, 1105 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{18}\text{H}_{19}\text{O}_3$ $[\text{M} + \text{H}]^+$ 283.1334, found 283.1328; R_f 0.38 (petroleum ether/EtOAc, 10:1).

4-(isochroman-1-yl)benzonitrile (3k). According to TP1, the reaction of isochroman (67 mg, 0.5 mmol) with (4-cyanophenyl) zinc reagent (7.1 mL, 0.21 M) afforded the desired product 3k, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:10) as a colorless oil (43 mg, 36%): ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.64 (d, J = 8.3 Hz, 2H), 7.43 (d, J = 8.3 Hz, 2H), 7.23–7.15 (m, 2H), 7.12–7.05 (m, 1H), 6.67 (d, J = 7.6 Hz, 1H), 5.76 (s, 1H), 4.17 (ddd, J = 11.2, 5.3, 3.8 Hz, 1H), 3.97–3.88 (m, 1H), 3.19–3.08 (m, 1H), 2.86–2.77 (m, 1H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 147.4, 135.9, 133.7, 132.3, 129.5, 129.1, 127.1, 126.5, 126.2, 118.7, 111.9, 78.8, 64.1, 28.6; IR (KBr) 3064, 2924, 2853, 2228, 1492, 1276, 1096 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{16}\text{H}_{14}\text{NO}$ $[\text{M} + \text{H}]^+$ 236.1075, found 236.1064; R_f 0.33 (petroleum ether/EtOAc, 10:1).

1-(Thiophen-2-yl)isochroman (3l). According to TP1, the reaction of isochroman (67 mg, 0.5 mmol) with thiophen-2-ylzinc reagent (6.0 mL, 0.25 M) afforded the desired product 3l, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:50) as a pale yellow solid (90 mg, 83%): mp 51–53 °C (ethyl ether/petroleum ether); ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.22–7.19 (m, 1H), 7.15–7.02 (m, 3H), 6.92–6.85 (m, 3H), 5.96 (s, 1H), 4.04 (ddd, J = 11.4, 6.3, 5.0 Hz, 1H), 3.84 (ddd, J = 11.4, 6.6, 4.9 Hz, 1H), 2.92–2.77 (m, 2H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 145.9, 136.3, 133.5, 128.8, 127.2, 126.9, 126.8, 126.2, 126.1, 125.8, 73.8, 62.3, 28.4; IR (KBr) 3138, 2961, 2873, 1603, 1491, 1401, 1284, 1108 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{13}\text{H}_{12}\text{NaOS}$ $[\text{M} + \text{Na}]^+$ 239.0507, found 239.0500; R_f 0.46 (petroleum ether/EtOAc, 20:1).

7-Methyl-1-(o-tolyl)isochroman (3m). According to TP1, the reaction of 7-methylisochroman (148 mg, 1 mmol) with 2-methylphenylzinc reagent (6.5 mL, 0.46 M) afforded the desired product 3m, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:50) as a colorless oil (148 mg, 62%): ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.24–7.03 (m, 5H), 6.99 (d, J = 7.8 Hz, 1H), 6.51 (s, 1H), 5.90 (s, 1H), 4.18 (ddd, J = 11.4, 5.5, 3.9 Hz, 1H), 3.90 (ddd, J = 11.4, 9.5, 3.9 Hz, 1H), 3.15–3.05 (m, 1H), 2.78 (dt, J = 16.2, 3.8 Hz, 1H), 2.37 (s, 3H), 2.19 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 139.9, 137.3, 137.2, 135.5, 130.9, 130.8, 129.9, 128.5, 127.9, 127.4, 126.7, 125.7, 77.6, 64.1, 28.5, 21.1, 19.4; IR (KBr) 3018, 2958, 2854, 1503, 1461, 1272, 1195, 1091 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{17}\text{H}_{19}\text{O}$ $[\text{M} + \text{H}]^+$ 239.1436, found 239.1430; R_f 0.56 (petroleum ether/EtOAc, 20:1).

7-Methoxy-1-(o-tolyl)isochroman (3n). According to TP1, the reaction of 7-methoxyisochroman (164 mg, 1 mmol) with 2-methylphenylzinc reagent (6.5 mL, 0.46 M) afforded the desired product 3n, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:20) as a colorless viscous oil (94 mg, 37%): ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.23–7.05 (m, 5H), 6.75 (dd, J = 8.3, 2.2 Hz, 1H), 6.23 (d, J = 2.5 Hz, 1H), 5.88 (s, 1H), 4.17 (ddd, J = 11.3, 5.4, 3.9 Hz, 1H), 3.89 (ddd, J = 11.3, 9.5, 3.9 Hz, 1H), 3.64 (s, 3H), 3.11–3.01 (m, 1H), 2.75 (dt, J = 15.9, 3.9 Hz, 1H), 2.35 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 157.8, 139.7, 138.5, 137.3, 130.9, 129.9, 129.5, 128.0, 126.2, 125.7, 112.6,

111.4, 77.8, 64.2, 55.2, 27.9, 19.4; IR (KBr) 3003, 2943, 2293, 1503, 1444, 1376, 1196, 1133, 1077 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{17}\text{H}_{18}\text{NaO}_2$ $[\text{M} + \text{Na}]^+$ 277.1204, found 277.1197; R_f 0.37 (petroleum ether/EtOAc, 10:1).

6,7-Dimethoxy-1-(o-tolyl)isochroman (3o). According to TP1, the reaction of 6,7-dimethoxyisochroman (194 mg, 1 mmol) with 2-methylphenylzinc reagent (6.5 mL, 0.46 M) afforded the desired product 3o, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:10) as a white solid (117 mg, 41%): mp 71–73 °C (ethyl ether/petroleum ether); ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.24–7.17 (m, 2H), 7.15–7.09 (m, 1H), 7.05 (d, J = 7.6 Hz, 1H), 6.65 (s, 1H), 6.19 (s, 1H), 5.87 (s, 1H), 4.12 (ddd, J = 11.3, 4.9, 4.8 Hz, 1H), 3.88 (s, 3H), 3.92–3.84 (m, 1H), 3.64 (s, 3H), 3.06–2.96 (m, 1H), 2.75 (dt, J = 16.1, 4.3 Hz, 1H), 2.38 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 147.8, 147.4, 139.7, 137.5, 130.9, 129.8, 129.1, 128.0, 126.2, 125.6, 111.3, 109.4, 77.2, 63.5, 55.9, 55.8, 28.3, 19.3; IR (KBr) 2957, 2929, 2861, 1609, 1518, 1463, 1375, 1245, 1126, 1078 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{18}\text{H}_{21}\text{O}_3$ $[\text{M} + \text{H}]^+$ 285.1491, found 285.1482; R_f 0.39 (petroleum ether/EtOAc, 5:1).

4-Methyl-1-(o-tolyl)isochroman (3p). According to TP1, the reaction of 4-methylisochroman (148 mg, 1 mmol) with 2-methylphenylzinc reagent (6.5 mL, 0.46 M) afforded the desired product 3p, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:50) as a colorless oil (153 mg, 64%): ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.31 (d, J = 7.8 Hz, 1H), 7.24–7.16 (m, 3H), 7.14–7.01 (m, 3H), 6.69 (d, J = 7.8 Hz, 1H), 5.96 (s, 1H), 4.12 (dd, J = 11.3, 5.1 Hz, 1H), 3.56 (dd, J = 11.3, 8.2 Hz, 1H), 3.17 (ddd, J = 19.8, 7.1, 6.9 Hz, 1H), 2.36 (s, 3H), 1.32 (d, J = 6.9 Hz, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 139.9, 139.3, 137.3, 136.8, 130.9, 129.9, 128.0, 126.8, 126.7, 126.2, 125.8, 125.6, 77.9, 69.9, 31.8, 19.4, 17.7; IR (KBr) 3022, 2959, 2868, 1489, 1462, 1381, 1133, 1086 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{17}\text{H}_{19}\text{O}$ $[\text{M} + \text{H}]^+$ 239.1436, found 239.1430; R_f 0.60 (petroleum ether/EtOAc, 20:1).

7-Bromo-1-(o-tolyl)isochroman (3q). According to TP1, the reaction of 7-bromoisochroman (213 mg, 1 mmol) with 2-methylphenylzinc reagent (6.5 mL, 0.46 M) afforded the desired product 3q, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:50) as a colorless oil (220 mg, 72%): ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.30 (dd, J = 8.3, 1.5 Hz, 1H), 7.24–7.11 (m, 3H), 7.09–7.01 (m, 2H), 6.83 (d, J = 1.5 Hz, 1H), 5.85 (s, 1H), 4.18 (ddd, J = 11.4, 5.6, 3.7 Hz, 1H), 3.89 (ddd, J = 11.4, 9.5, 4.0 Hz, 1H), 3.12–3.01 (m, 1H), 2.77 (ddd, J = 16.5, 3.8, 3.7 Hz, 1H), 2.33 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 139.7, 138.9, 137.2, 132.9, 131.1, 130.3, 129.9, 129.7, 129.1, 128.4, 125.9, 119.7, 77.4, 63.8, 28.3, 19.4; IR (KBr) 3019, 2925, 2853, 1593, 1482, 1266, 1175, 1090 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{16}\text{H}_{13}\text{BrNaO}$ $[\text{M} + \text{Na}]^+$ 325.0204, found 325.0194; R_f 0.43 (petroleum ether/EtOAc, 20:1).

5-Bromo-1-(o-tolyl)isochroman (3r). According to TP1, the reaction of 5-bromoisochroman (213 mg, 1 mmol) with 2-methylphenylzinc reagent (6.5 mL, 0.46 M) afforded the desired product 3r, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:50) as a white solid (187 mg, 62%): mp 98–100 °C (ethyl ether/petroleum ether); ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.44 (d, J = 7.8 Hz, 1H), 7.24–7.10 (m, 3H), 7.05 (d, J = 7.3 Hz, 1H), 6.94 (t, J = 7.8 Hz, 1H), 6.66 (d, J = 7.8 Hz, 1H), 5.87 (s, 1H), 4.22 (ddd, J = 11.6, 5.8, 3.9 Hz, 1H), 3.92 (ddd, J = 11.6, 9.2, 4.5 Hz, 1H), 3.07–2.96 (m, 1H), 2.88 (dt, J = 17.1, 4.0 Hz, 1H), 2.33 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 140.2, 139.3, 137.3, 133.8, 131.0, 130.5, 129.9, 128.3, 127.2, 125.8, 125.1, 77.6, 63.8, 29.6, 19.4; IR (KBr) 3068, 2984, 2921, 1559, 1490, 1443, 1380, 1315, 1256, 1172, 1095, 1053 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{16}\text{H}_{13}\text{BrNaO}$ $[\text{M} + \text{Na}]^+$ 325.0204, found 325.0196; R_f 0.54 (petroleum ether/EtOAc, 20:1).

7-Chloro-1-(o-tolyl)isochroman (3s). According to TP1, the reaction of 7-chloroisochroman (169 mg, 1 mmol) with 2-methylphenylzinc reagent (6.5 mL, 0.46 M) afforded the desired product 3s, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:50) as a colorless oil (168 mg, 65%): ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.24–7.04 (m, 6H), 6.68 (d, J =

1.7 Hz, 1H), 5.85 (s, 1H), 4.19 (ddd, $J = 11.4, 5.6, 3.7$ Hz, 1H), 3.89 (ddd, $J = 11.4, 9.6, 3.9$ Hz, 1H), 3.14–3.04 (m, 1H), 2.78 (dt, $J = 16.5, 3.8$, 1H), 2.33 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 139.4, 139.0, 137.2, 132.4, 131.7, 131.1, 130.0, 129.8, 128.3, 126.8, 126.2, 125.9, 77.5, 63.9, 28.2, 19.4; IR (KBr) 3021, 2962, 2853, 1599, 1484, 1267, 1177, 1091 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{16}\text{H}_{15}\text{ClNaO}$ [$\text{M} + \text{Na}$] $^+$ 281.0709, found 281.0701; R_f 0.41 (petroleum ether/EtOAc, 20:1).

1-(*o*-Tolyl)isothiochroman (3t). According to TP1, the reaction of isothiochroman (150 mg, 1 mmol) with 2-methylphenylzinc reagent (6.5 mL, 0.46 M) afforded the desired product 3t, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:50) as a pale yellow oil (132 mg, 55%): ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.23–7.11 (m, 4H), 7.10–7.04 (m, 2H), 6.84 (dd, $J = 11.9, 7.7$ Hz, 2H), 5.33 (s, 1H), 3.24 (ddd, $J = 16.3, 5.1, 5.0$ Hz, 1H), 3.19–3.10 (m, 1H), 2.93–2.83 (m, 2H), 2.45 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 140.8, 136.9, 136.8, 136.2, 130.7, 129.8, 129.2, 128.9, 126.9, 126.8, 126.1, 125.7, 42.4, 31.1, 24.3, 19.7; IR (KBr) 2963, 2920, 2360, 1645, 1486, 1261, 1196, 1132 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{16}\text{H}_{17}\text{S}$ [$\text{M} + \text{H}$] $^+$ 241.1051, found 241.1046; R_f 0.62 (petroleum ether/EtOAc, 20:1).

9-(*o*-Tolyl)-9H-xanthene (3u). According to TP1, the reaction of 9H-xanthene (182 mg, 1 mmol) with 2-methylphenylzinc reagent (6.5 mL, 0.46 M) afforded the desired product 3u, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:100) as a white solid (72 mg, 26%): mp 129–131 $^{\circ}\text{C}$ (ethyl ether/petroleum ether); ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.23–7.15 (m, 6H), 7.10 (dd, $J = 8.1, 1.2$ Hz, 2H), 6.93 (td, $J = 7.4, 1.4$ Hz, 2H), 6.87 (d, $J = 7.6$ Hz, 2H), 5.56 (s, 1H), 2.22 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 150.9, 143.5, 135.9, 131.2, 131.1, 129.2, 127.8, 126.9, 126.4, 124.1, 123.1, 116.3, 41.3, 20.0; IR (KBr) 3015, 2925, 1599, 1571, 1461, 1449, 1321, 1254, 1120, 1034 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{20}\text{H}_{16}\text{NaO}$ [$\text{M} + \text{Na}$] $^+$ 295.1099, found 295.1092; R_f 0.60 (petroleum ether/EtOAc, 50:1).

1-(Methoxy(4-methoxyphenyl)methyl)-2-methylbenzene (3v). According to TP1, the reaction of 1-methoxy-4-(methoxymethyl)-benzene (76 mg, 0.5 mmol) with 2-methylphenylzinc reagent (3.3 mL, 0.46 M) afforded the desired product 3v, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:20) as a colorless oil (56 mg, 45%): ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.44 (d, $J = 7.3$ Hz, 1H), 7.24–7.10 (m, 5H), 6.84 (d, $J = 8.8$ Hz, 2H), 5.35 (s, 1H), 3.78 (s, 3H), 3.36 (s, 3H), 2.23 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 158.9, 139.8, 135.8, 133.1, 130.5, 128.9, 127.3, 126.5, 125.9, 113.7, 82.2, 56.9, 55.2, 19.3; IR (KBr) 2931, 2835, 1610, 1510, 1461, 1303, 1247, 1172, 1096 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{16}\text{H}_{18}\text{NaO}_2$ [$\text{M} + \text{Na}$] $^+$ 265.1204, found 265.1196; R_f 0.65 (petroleum ether/EtOAc, 10:1).

1-((4-Bromophenyl)(methoxymethyl)-2-methylbenzene (3w). According to TP1, the reaction of 1-bromo-4-(methoxymethyl)-benzene (201 mg, 1 mmol) with 2-methylphenylzinc reagent (6.5 mL, 0.46 M) afforded the desired product 3w, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:80) as a colorless oil (149 mg, 51%): ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.35 (d, $J = 8.6$ Hz, 2H), 7.31–7.26 (m, 1H), 7.16–7.03 (m, 5H), 5.27 (s, 1H), 3.28 (s, 3H), 2.16 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 140.1, 138.9, 135.9, 131.4, 130.7, 129.1, 127.7, 126.9, 126.1, 121.3, 82.0, 57.0, 19.4; IR (KBr) 2935, 2822, 1660, 1588, 1484, 1396, 1281, 1092 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{15}\text{H}_{15}\text{BrNaO}$ [$\text{M} + \text{Na}$] $^+$ 313.0204, found 313.0193; R_f 0.38 (petroleum ether/EtOAc, 50:1).

Methyl(phenyl(*o*-tolyl)methyl)sulfane (3x). According to TP1, the reaction of benzyl methyl sulfide (138 mg, 1 mmol) with 2-methylphenylzinc reagent (6.5 mL, 0.46 M) afforded the desired product 3x, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:100) as a colorless oil (96 mg, 42%): ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.58 (d, $J = 7.6$ Hz, 1H), 7.39–7.26 (m, 4H), 7.24–7.11 (m, 4H), 5.24 (s, 1H), 2.33 (s, 3H), 2.00 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 140.6, 138.9, 136.1, 130.5, 128.6, 128.4, 128.1, 127.0, 126.9, 126.2, 52.4, 19.6, 16.1; IR (KBr) 2953, 2852, 1491, 1451, 1261, 1133, 1076 cm^{-1} ; HRMS (ESI)

calcd for $\text{C}_{15}\text{H}_{16}\text{NaS}$ [$\text{M} + \text{Na}$] $^+$ 251.0870, found 251.0864; R_f 0.55 (petroleum ether/EtOAc, 50:1).

6-Bromo-2-(*o*-tolyl)-2H-chromene (3y). According to TP1, the reaction of 6-bromo-2H-chromene (105 mg, 0.5 mmol) with 2-methylphenylzinc reagent (3.3 mL, 0.46 M) afforded the desired product 3y, which was isolated by flash chromatography (eluent ethyl acetate/petroleum ether = 1:100) as a colorless oil (102 mg, 68%): ^1H NMR (400 MHz, CDCl_3) δ (ppm) 7.41 (d, $J = 7.3$ Hz, 1H), 7.24–7.10 (m, 5H), 6.62 (d, $J = 8.6$ Hz, 1H), 6.49 (dd, $J = 9.8, 1.9$ Hz, 1H), 6.16–6.12 (m, 1H), 5.79 (dd, $J = 9.8, 3.3$ Hz, 1H), 2.45 (s, 3H); ^{13}C NMR (100 MHz, CDCl_3) δ (ppm) 152.4, 137.7, 136.0, 131.9, 130.9, 129.0, 128.5, 127.7, 126.2, 125.6, 123.5, 123.2, 117.7, 113.0, 74.8, 19.2; IR (KBr) 3018, 2927, 2861, 1494, 1452, 1403, 1059 cm^{-1} ; HRMS (ESI) calcd for $\text{C}_{16}\text{H}_{13}\text{BrNaO}$ [$\text{M} + \text{Na}$] $^+$ 323.0047, found 323.0036; R_f 0.58 (petroleum ether/EtOAc, 50:1).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.joc.8b00776.

^1H NMR and ^{13}C NMR spectra of compounds (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: zpech@upc.edu.cn.

ORCID

Zhihua Peng: 0000-0001-6971-4456

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We thank the Shandong Provincial Natural Science Foundation (ZR2012BQ006), the National Natural Science Foundation of China (21202205), and the Fundamental Research Funds for the Central Universities (17CX02069) for financial support.

■ REFERENCES

- (a) Markaryan, E. A.; Samodurova, A. G. Advances in the chemistry of isochroman. *Russ. Chem. Rev.* **1989**, *58*, 479–493. (b) Yamaori, S.; Kushiara, M.; Yamamoto, I.; Watanabe, K. Characterization of Major Phytocannabinoids, Cannabidiol and Cannabinol, as Isoform-Selective and Potent Inhibitors of Human CYP1 Enzymes. *Biochem. Pharmacol.* **2010**, *79*, 1691–1698. (c) Albrecht, U.; Lalk, M.; Langer, P. Synthesis and Structure–activity Relationships of 2-Vinylchroman-4-Ones as Potent Antibiotic Agents. *Bioorg. Med. Chem.* **2005**, *13*, 1531–1536. (d) Ameen, D.; Snape, T. J. Chiral 1,1-Diaryl Compounds as Important Pharmacophores. *MedChemComm* **2013**, *4*, 893–907.
- (a) Lu, Z.-Y.; Lin, Z.-J.; Wang, W.-L.; Du, L.; Zhu, T.-J.; Fang, Y.-C.; Gu, Q.-Q.; Zhu, W.-M. Citrinin Dimers from the Halotolerant Fungus *Penicillium Citrinum* B-57. *J. Nat. Prod.* **2008**, *71*, 543–546. (b) de Groot, M. J.; Alex, A. A.; Jones, B. C. Development of a Combined Protein and Pharmacophore Model for Cytochrome P450 2C9. *J. Med. Chem.* **2002**, *45*, 1983–1993. (c) Clark, B. R.; Capon, R. J.; Lacey, E.; Tennant, S.; Gill, J. H. Citrinin Revisited: from Monomers to Dimers and beyond. *Org. Biomol. Chem.* **2006**, *4*, 1520–1528. (d) Liu, H.-C.; Du, L.; Zhu, T.-J.; Li, D.-H.; Geng, M. Y.; Gu, Q.-Q. Two New Citrinin Dimers from a Volcano Ash-Derived Fungus, *Penicillium citrinum* HG1–5. *Helv. Chim. Acta* **2010**, *93*, 2224–2230. (e) Silvestri, R.; Artico, M.; De Martino, G.; Ragno, R.; Massa, S.; Loddo, R.; Murgioni, C.; Loi, A. G.; La Colla, P.; Pani, A. Synthesis, Biological Evaluation, and Binding Mode of Novel 1-[2-(Diarylmethoxy)ethyl]-2-Methyl-5-Nitroimidazoles Targeted at the HIV-1 Reverse Transcriptase. *J. Med. Chem.* **2002**, *45*, 1567–1576.

- (3) For selected examples, see: (a) Maier, C. A.; Wünsch, B. Novel Spiropiperidines as Highly Potent and Subtype Selective σ -Receptor Ligands. Part 1. *J. Med. Chem.* **2002**, *45*, 438–448. (b) De Martino, G.; La Regina, G.; Di Pasquali, A.; Ragno, R.; Bergamini, A.; Ciaprin, C.; Sinistro, A.; Maga, G.; Crespan, E.; Artico, M.; Silvestri, R. Novel 1-[2-(Diarylmethoxy)Ethyl]-2-Methyl-5-Nitroimidazoles as HIV-1 Non-Nucleoside Reverse Transcriptase Inhibitors. A Structure-Activity Relationship Investigation. *J. Med. Chem.* **2005**, *48*, 4378–4388. (c) Lehmann, F.; Pettersen, A.; Currier, E. A.; Sherbukhin, V.; Olsson, R.; Hacksell, U.; Luthman, K. Novel Potent and Efficacious Nonpeptidic Urotensin II Receptor Agonists. *J. Med. Chem.* **2006**, *49*, 2232–2240.
- (4) For selected references, see: (a) Liu, C.; Zhang, H.; Shi, W.; Lei, A. Bond Formations between Two Nucleophiles: Transition Metal Catalyzed Oxidative Cross-Coupling Reactions. *Chem. Rev.* **2011**, *111*, 1780–1824. (b) Rohlmann, R.; García Mancheño, O. Metal-Free Oxidative C(sp³)-H Bond Couplings as Valuable Synthetic Tools for C–C Bond Formations. *Synlett* **2013**, *24*, 6–10. (c) Schweitzer-Chaput, B.; Sud, A.; Pintér, Á.; Dehn, S.; Schulze, P.; Klusmann, M. Synergistic Effect of Ketone and Hydroperoxide in Brønsted Acid Catalyzed Oxidative Coupling Reactions. *Angew. Chem., Int. Ed.* **2013**, *52*, 13228–13232. (d) Meng, Z.; Sun, S.; Yuan, H.; Lou, H.; Liu, L. Catalytic Enantioselective Oxidative Cross-Coupling of Benzylic Ethers with Aldehydes. *Angew. Chem., Int. Ed.* **2014**, *53*, 543–547. (e) Li, Z.; Yu, R.; Li, H. Iron-Catalyzed C–C Bond Formation by Direct Functionalization of C–H Bonds Adjacent to Heteroatoms. *Angew. Chem., Int. Ed.* **2008**, *47*, 7497–7500.
- (5) For selected reviews, see: (a) Yeung, C. S.; Dong, V. M. Catalytic Dehydrogenative Cross-Coupling: Forming Carbon–Carbon Bonds by Oxidizing Two Carbon–Hydrogen Bonds. *Chem. Rev.* **2011**, *111*, 1215–1292. (b) Girard, S. A.; Knauber, T.; Li, C.-J. The Cross-Dehydrogenative Coupling of C_{sp}³-H Bonds: A Versatile Strategy for C–C Bond Formations. *Angew. Chem., Int. Ed.* **2014**, *53*, 74–100. (c) Li, C.-J. Cross-Dehydrogenative Coupling (CDC): Exploring C–C Bond Formations beyond Functional Group Transformations. *Acc. Chem. Res.* **2009**, *42*, 335–344. (d) Roizen, J. L.; Harvey, M. E.; Du Bois, J. Metal-Catalyzed Nitrogen–Atom Transfer Methods for the Oxidation of Aliphatic C–H Bonds. *Acc. Chem. Res.* **2012**, *45*, 911–922. (e) Varun, B. V.; Dhineshkumar, J.; Bettadapur, K. R.; Siddaraju, Y.; Alagiri, K.; Prabhu, K. R. Recent Advancements in Dehydrogenative Cross Coupling Reactions for C–C Bond Formation. *Tetrahedron Lett.* **2017**, *58*, 803–824.
- (6) For selected references, see: (a) Zhang, Y.; Li, C.-J. DDQ-Mediated Direct Cross-Dehydrogenative-Coupling (CDC) between Benzylic Ethers and Simple Ketones. *J. Am. Chem. Soc.* **2006**, *128*, 4242–4243. (b) Zhang, Y.; Li, C.-J. Highly Efficient Cross-Dehydrogenative-Coupling between Ethers and Active Methylene Compounds. *Angew. Chem., Int. Ed.* **2006**, *45*, 1949–1952. (c) Ghobrial, M.; Harhammer, K.; Mihovilovic, M. D.; Schnürch, M. Facile, Solvent and Ligand Free Iron Catalyzed Direct Functionalization of N-Protected Tetrahydroisoquinolines and Isochroman. *Chem. Commun.* **2010**, *46*, 8836–8838. (d) Liu, X.; Sun, B.; Xie, Z.; Qin, X.; Liu, L.; Lou, H. Manganese Dioxide-Methanesulfonic Acid Promoted Direct Dehydrogenative Alkylation of sp³ C–H Bonds Adjacent to a Heteroatom. *J. Org. Chem.* **2013**, *78*, 3104–3112. (e) Pan, X.; Hu, Q.; Chen, W.; Liu, X.; Sun, B.; Huang, Z.; Zeng, Z.; Wang, L.; Zhao, D.; Ji, M.; Liu, L.; Lou, H. Copper(II) Catalyzed Cross-Dehydrogenative Coupling of Cyclic Benzylic Ethers with Simple Carbonyl Compounds by Na₂S₂O₈. *Tetrahedron* **2014**, *70*, 3447–3451. (f) Feng, J.; Lv, M. F.; Lu, G. P.; Cai, C. Selective Formation of C–N and C = N Bonds via C(sp³)-H Activation of Isochroman in the Presence of DTBP. *Org. Chem. Front.* **2015**, *2*, 60–64.
- (7) For selected examples, see: (a) Qvortrup, K.; Rankic, D. A.; MacMillan, D. W. C. A General Strategy for Organocatalytic Activation of C–H Bonds via Photoredox Catalysis: Direct Arylation of Benzylic Ethers. *J. Am. Chem. Soc.* **2014**, *136*, 626–629. (b) Yang, Q.; Li, S.; Wang, J. Cobalt-Catalyzed Cross-Dehydrogenative Coupling of imidazo[1,2-*a*]pyridines with Isochroman Using Molecular Oxygen as the Oxidant. *Org. Chem. Front.* **2018**, *5*, 577–581. (c) Park, S. J.; Price, J. R.; Todd, M. H. Oxidative Arylation of Isochroman. *J. Org. Chem.* **2012**, *77*, 949–955. (d) Ghobrial, M.; Schnürch, M.; Mihovilovic, M. D. Direct Functionalization of (Un)protected Tetrahydroisoquinoline and Isochroman under Iron and Copper Catalysis: Two Metals, Two Mechanisms. *J. Org. Chem.* **2011**, *76*, 8781–8793.
- (8) (a) Muramatsu, W.; Nakano, K.; Li, C.-J. Simple and Direct sp³ C–H Bond Arylation of Tetrahydroisoquinolines and Isochromans via 2,3-Dichloro-5,6-Dicyano-1,4-Benzoquinone Oxidation under Mild Conditions. *Org. Lett.* **2013**, *15*, 3650–3653. (b) Muramatsu, W.; Nakano, K. Organocatalytic Approach for C(sp³)-H Bond Arylation, Alkylation, and Amidation of Isochromans under Facile Conditions. *Org. Lett.* **2014**, *16*, 2042–2045. (c) Muramatsu, W.; Nakano, K. Efficient C(sp³)-H Bond Functionalization of Isochroman by AZADOL Catalysis. *Org. Lett.* **2015**, *17*, 1549–1552. (d) Muramatsu, W.; Nakano, K. Hypervalent iodine(III)-Mediated C(sp³)-H Bond Arylation, Alkylation, and Amidation of Isothiochroman. *Tetrahedron Lett.* **2015**, *56*, 437–440.
- (9) Chen, W.; Xie, Z.; Zheng, H.; Lou, H.; Liu, L. Structurally Diverse α -Substituted Benzopyran Synthesis through a Practical Metal-Free C(sp³)-H Functionalization. *Org. Lett.* **2014**, *16*, 5988–5991.
- (10) (a) Haag, B.; Mosrin, M.; Ila, H.; Malakhov, V.; Knochel, P. Regio- and Chemoselective Metalation of Arenes and Heteroarenes Using Hindered Metal Amide Bases. *Angew. Chem., Int. Ed.* **2011**, *50*, 9794–9824. (b) Benischke, A. D.; Ellwart, M.; Becker, M. R.; Knochel, P. Polyfunctional Zinc and Magnesium Organometallics for Organic Synthesis: Some Perspectives. *Synthesis* **2016**, *48*, 1101–1107. (c) Knochel, P.; Schade, M. A.; Bernhardt, S.; Manolikakes, G.; Metzger, A.; Piller, F. M.; Rohbogner, C. J.; Mosrin, M. Functionalization of Heterocyclic Compounds Using Polyfunctional Magnesium and Zinc Reagents. *Beilstein J. Org. Chem.* **2011**, *7*, 1261–1277.
- (11) Wang, T.; Schrempp, M.; Berndhäuser, A.; Schiemann, O.; Menche, D. Efficient and General Aerobic Oxidative Cross-Coupling of THIQs with Organozinc Reagents Catalyzed by CuCl₂: Proof of a Radical Intermediate. *Org. Lett.* **2015**, *17*, 3982–3985.
- (12) Peng, Z.; Yu, Z.; Chen, D.-H.; Liang, S.; Zhang, L.; Zhao, D.; Song, L.; Jiang, C. Efficient C(sp³)-H Bond Arylation of Tetrahydroisoquinolines with Knochel-Type Arylzinc Reagents under Oxidative Conditions. *Synlett* **2017**, *28*, 1835–1839.
- (13) (a) Piller, F. M.; Appukkuttan, P.; Gavryushin, A.; Helm, M.; Knochel, P. Convenient Preparation of Polyfunctional Aryl Magnesium Reagents by a Direct Magnesium Insertion in the Presence of LiCl. *Angew. Chem., Int. Ed.* **2008**, *47*, 6802–6806. (b) Metzger, A.; Bernhardt, S.; Manolikakes, G.; Knochel, P. MgCl₂-Accelerated Addition of Functionalized Organozinc Reagents to Aldehydes, Ketones, and Carbon Dioxide. *Angew. Chem., Int. Ed.* **2010**, *49*, 4665–4668. (c) Buesking, A. W.; Baguley, T. D.; Ellman, J. A. Asymmetric Synthesis of Amines by the Knochel-Type MgCl₂-Enhanced Addition of Benzyl Zinc Reagents to *N*-Tert-Butanesulfinyl Aldimines. *Org. Lett.* **2011**, *13*, 964.
- (14) Krasovskiy, A.; Malakhov, V.; Gavryushin, A.; Knochel, P. Efficient Synthesis of Functionalized Organozinc Compounds by the Direct Insertion of Zinc into Organic Iodides and Bromides. *Angew. Chem., Int. Ed.* **2006**, *45*, 6040–6044.
- (15) (a) Tsang, A. S.-K.; Jensen, P.; Hook, J. M.; Hashmi, A. S. K.; Todd, M. H. An Oxidative Carbon–carbon Bond-Forming Reaction Proceeds via an Isolable Iminium Ion. *Pure Appl. Chem.* **2011**, *83*, 655–665. (b) Jung, H. H.; Floreancig, P. E. Mechanistic Analysis of Oxidative C–H Cleavages Using Inter- and Intramolecular Kinetic Isotope Effects. *Tetrahedron* **2009**, *65*, 10830–10836.